

Coding & Solution Evaluation Document

Date	15 March 2026	Team ID	SWTID-2026-6684
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models		
Maximum Marks	5 Marks		

Coding & Solution Description: This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

INSTRUCTIONS:

- 1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
- 2. The solution should address the core problem statement and implement the key functional requirements.
- 3. Include all source files, configuration files, and setup instructions needed to run the application.
- 4. Attach or link to your code repository (e.g., GitHub) in the table below.

SOLUTION SUMMARY

FIELD	DETAILS				
Repository Link / URL	https://github.com/SmartBridge-SWT/ComicCraft-Gemini-AI				
Programming Language(s)	Python 3.11+, JavaScript (ES6+), HTML5, CSS3				
Framework(s) Used	FastAPI / Streamlit (Backend & UI), Google GenAI SDK, React / TailwindCSS				
Key Features Implemented		Automated Script Generation: Utilizes Gemini 1.5 Pro to synthesize user prompts into multi-panel comic scripts, dialogue, and character descriptions.	Visual Panel Rendering: Integrates Imagen / Gemini visual multimodal capabilities to generate consistent panel artwork.	Dynamic Layout Engine: Organizes panels, speech bubbles, and text overlays into printable multi-page PDF/ image comic books.	Interactive Editing Workflow: Allows real-time dialogue adjustments, character prompt tuning, and scene regenerations.
		Multi-user collaborative editing in	Direct export integrations		

FIELD	DETAILS		
Pending / Incomplete Features		real-time via WebSockets.	to third-party social media platforms and webcomic publishing APIs.
Setup / Run Instructions	<pre># 1. Clone repository git clone https://github.com/SmartBridge-SWT/ComicCraft-Gemini-AI.git cd ComicCraft-Gemini-AI # 2. Set up virtual environment & install requirements python -m venv venv && source venv/bin/activate pip install -r requirements.txt # 3. Configure API Keys (.env) GEMINI_API_KEY="your_google_gemini_api_key_here" # 4. Launch Application streamlit run app.py # or uvicorn main:app --reload</pre>		

CODE QUALITY CHECKLIST

S.NO	CRITERIA	STATUS (YES / NO)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

ADDITIONAL NOTES / COMMENTS

The application architecture adheres to modern clean code standards, utilizing modular services for Gemini API call orchestration, image asset caching, and layout PDF rendering. Comprehensive unit testing and exception handling wrappers guarantee continuous runtime stability.

